Saigon Rising

A turn-based planning game about keeping a riverside district of Hồ Chí Minh City dry from 2026 to 2050

RMIT GenAI & Cybersecurity Hackathon 2025 · Challenge 3: Vibe Coding, Play to Impact · Team **Quadcore**



The planning screen: the district map with block heights shown, the tide gauge and forecast, and the tool tray.

1. Introduction

Saigon Rising puts the player in charge of Bến Nước, a fictional district on the bank of the Saigon River. Every two years the rainy season arrives. The player reads a forecast of rain and peak tide, spends a limited budget on flood defences, and then watches the season play out on the map: rain falls, the tide pushes in from the river and up the canal, and water runs downhill into the lowest streets. A report shows how many residents were flooded, what it cost, and one real fact about flooding in the city.

After twelve seasons, from 2026 to 2050, the player receives a grade based on how many residents stayed dry. If repeated flooding drives residents' happiness to zero first, the city replaces the planner. The game runs entirely in the browser with HTML, CSS and JavaScript, and was built with Claude (Anthropic) as the AI coding partner.

2. Why flooding in Hồ Chí Minh City

Flooding is one of the most visible everyday problems in Hồ Chí Minh City, and it is getting worse. We chose it because it is local to our campus, it affects almost everyone who lives here, and its causes are widely misunderstood: many people blame rain alone, when the real story combines several factors that make each other worse.

- **The city is low.** Nearly half of the city's land is less than one metre above sea level [1].

- **Tides and rain combine.** Major floods happen when heavy rain coincides with high tides, so rivers and canals spill onto roads and rainwater cannot drain out [1][4].
- **The ground is sinking.** Researchers estimate the city sinks about 2–5 cm a year on average, and 7–8 cm a year in some densely built areas — more than twice the rate of sea-level rise. Falling groundwater levels and rapid urbanisation are among the main causes [2].
- **The drainage system has not kept up.** Much of the canal and drainage network dates from a far smaller city, and fast growth has outpaced it [1].
- **Nature helps.** The Cần Giờ mangrove forest, recognised as a UNESCO biosphere reserve in 2000, reduces the impact of rising seas on the city [3].
- **People are affected unequally.** Low-lying neighbourhoods and small businesses are hit hardest; at high tide, water can reach knee height in some streets [1].

In 2026 the city adopted a climate action plan to 2030 that upgrades drainage and canals and expands green space, alongside a large plan to fix flood hot spots [1]. A game is a good way to make these trade-offs tangible: it lets players feel why a single fix rarely works.

3. Potential impact

The game is designed for secondary-school and university students and for anyone who lives with the rainy season but has never thought about why some streets flood and others do not. Its impact comes from three design choices:

- **Every mechanic is a real lesson.** Players discover by failing that drains are useless when their outlets are underwater, that the land keeps sinking until groundwater pumping stops, and that selling wetlands to developers floods the neighbours.
- **Short sessions.** A full game takes about ten minutes, so it fits in a classroom lesson or a commute.
- **Replay with purpose.** Weather is randomised each game and the end screen explains what worked and what did not, inviting players to try a different plan.

The same approach could support community workshops on flood planning, or be translated into Vietnamese for local schools. Because it needs only a browser and no installation, it can be shared with a single link.

4. Game mechanics

Each round follows the same loop: **forecast → build → season → report**. The map is a 10 × 7 grid of blocks, each with a height above the river. Homes, apartments and the school hold residents; the market pays the most tax. A block under more than 15 cm of water counts as flooded.

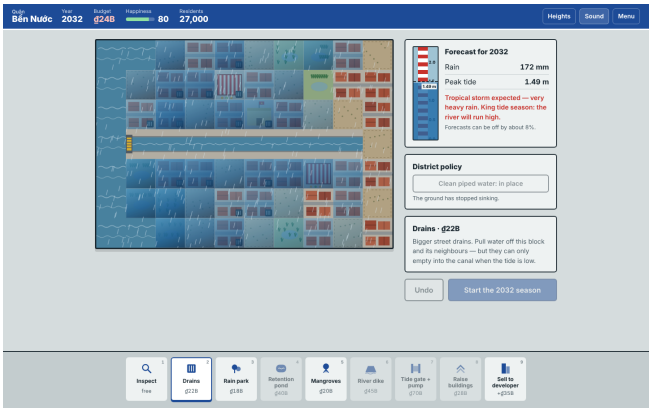
Tool	Cost	Effect and the lesson behind it
Drains	₫22B	Remove water from a block and its neighbours — but at high tide the outlets are underwater, so they work at only 20–50%.
Rain park	₫18B	Turns an empty lot into ground that sheds 25% of its rain instead of 60%.
Retention pond	₫40B	A basin that collects up to 0.9 m of water from surrounding blocks (sponge-city design).
Mangroves	₫20B	Add 35 cm of protection against the tide on the riverbank.
River dike	₫45B	Adds 1 m of protection — but also stops rainwater draining back to the river.
Tide gate + pump	₫70B	At the canal mouth: blocks the tide from entering the canal and keeps every drain working at full power.
Raise buildings	₫28B	Lifts a home, market or school by 40 cm (twice at most).
Sell to developer	+₫35B	Money now and more tax later, but apartments shed 95% of their rain.
Clean piped water	₫90B	A one-off policy that stops groundwater pumping, so the land stops sinking 4 cm per season.

The season simulation

When a season starts, the tide floods outward from the river into every connected block lower than the tide, and up the canal unless it is gated. Rain then adds water according to each block's surface: concrete sheds almost all of it, wetland very little. Water flows from higher to lower water surfaces over several passes; riverbank blocks without a dike drain back to the river, and the canal mouth empties only down to the tide level unless a gate and pump are built. Drains then remove water, scaled by how well their outlets can empty at that tide. After each season the whole district sinks 4 cm until the clean-water policy is bought, while sea level and storm intensity rise over the decades.

Money and happiness

Dry blocks pay full tax; flooded ones pay 40% and cost repairs. Happiness rises in dry seasons and with green space, and falls with the share of residents flooded. This creates the central tension: defences cost money, but floods cost both money and goodwill.



A tropical storm meets a king tide in 2032: the tide has flooded the riverbank while rain pools in the low streets.

5. Technology stack

- **HTML, CSS and plain JavaScript**, with no frameworks, backend or build step. The scripts are classic <script> files rather than ES modules, so the game also runs when index.html is opened straight from disk.
- **Inline SVG** for every tile, overlay and icon, generated in code. No image files are needed, and the art scales sharply to any screen size.
- **Canvas** for the rain animation, and **CSS transitions** for the rising tide gauge and floodwater.
- **Web Audio API** for all sound effects (rain, thunder, clicks, chimes), synthesised in code.
- **Be Vietnam Pro** from Google Fonts, a typeface designed for Vietnamese diacritics, with a system-font fallback offline.
- **Claude (Anthropic), Claude Opus 5.5** as the AI partner for concept, visual design, code, testing and fact-checking. The prompts are recorded in the prompts/ folder.
- **Testing tools used by the AI:** Node.js to run the simulation headlessly, and Playwright with Chromium to play the game automatically and capture screenshots.

The code separates the rules (data.js, sim.js) from the interface (art.js, audio.js, main.js). Because sim.js contains no page code, the same rules could be tested in Node by simulating hundreds of games.

6. How we used AI, and how we tested it

We chose the issue and the game style from options the AI proposed, then had it plan the mechanics and a visual identity before writing any code. Instead of trusting the first version, we tested it the way a designer would:

- **Balance testing.** 300 simulated games for each of four strategies (do nothing, sell everything, drains only, a sensible plan). This showed that the first season was too punishing for new players, and later that taxes were so generous money stopped mattering (₫770B left unspent). After two rounds of tuning, careless strategies always fail and a sensible plan earns mostly a B, finishing with about ₫46B — so every purchase is a real choice.
- **Visual review.** Automated playthroughs captured screenshots of every screen on desktop and phone. These revealed a flat blue box where the half-submerged title should be, an overlapping gauge label, faint floodwater, and a tool tray that covered half a phone screen. All were fixed.
- **Fact-checking.** The AI's first draft of the in-game facts was broadly right but vague ("a metre or two above sea level"). Checking sources made them precise: nearly half the city is under one metre, and it sinks 2–5 cm a year.

7. Reflection

What worked. Designing the mechanics around real causes, rather than adding facts on top of a generic game, made the educational content part of the gameplay. Players learn about tidal blocking because their drains stop working, not because a pop-up tells them. Asking the AI for a design plan before code also produced a far more distinctive look than a direct "make a game" prompt would.

Where AI needed checking. The AI produced working code quickly, but its first balance was wrong in both directions, and only systematic testing revealed it. Its factual claims also needed sources. Our main lesson is that vibe coding is fast at producing a first version, but a good result depends on testing, measuring and questioning that version.

Fundamentals still matter. We have also learnt that we need to understand the fundamentals of coding before using vibe coding, to get the best and most satisfactory results. Knowing how HTML,

CSS and JavaScript fit together is what let us read the code the AI produced, spot when something was wrong, and give it clear instructions to fix it. Vibe coding made us faster, but it did not replace understanding what we were building.

What we would do next. Add a Vietnamese-language version, more districts with different geography (such as an inner-city district with no river frontage), a sandbox mode with fixed weather for classroom use, and real elevation data for an actual neighbourhood.

Sources

[1] "Vietnam's Ho Chi Minh City learning to live with floods but climate change has raised the stakes", The Star (Associated Press), 1 October 2026. https://www.thestar.com.my/aseanplus/aseanplus-news/2026/10/01/vietnam039s-ho-chi-minh-city-learning-to-live-with-floods-but-climate-change-has-raised-the-stakes

[2] "Ho Chi Minh City's severe land subsidence heightens flood risks: study", Tuoi Tre News, 15 September 2025. https://news.tuoitre.vn/ho-chi-minh-citys-severe-land-subsidence-heightens-flood-risks-study-103250915163107199.htm

[3] "Can Gio Mangrove Biosphere Reserve, Ho Chi Minh City", Vietnam Nature Conservation and Biodiversity Agency. https://en.nbca.gov.vn/khu-du-tru-sinh-quyen-rung-ngap-man-can-gio-tp-ho-chi-minh-khu-dtsq-can-gio/

[4] "Vietnam's Ho Chi Minh City, Mekong Delta face highest tides of 2026", The Star, 29 September 2026. https://www.thestar.com.my/aseanplus/aseanplus-news/2026/09/29/vietnam039s-ho-chi-minh-city-mekong-delta-face-highest-tides-of-2026